Chapter 6 — Chemical Energetics

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. ΔH equals:

(a) H reactants - H products **(b)** Mass products **(c)** H products - H reactants **(d)** Temperature change

2. Exothermic reaction has:

(a) Positive ΔH **(b)** Negative ΔH **(c)** Zero ΔH always **(d)** No heat

3. Endothermic reaction has:

(a) Positive ΔH **(b)** Negative ΔH **(c)** Zero mass **(d)** No bonds

4. Activation energy is shown by:

(a) Initial level **(b)** Final level **(c)** Profile peak **(d)** Zero line

5. Catalyst changes:

(a) ΔH **(b)** Equilibrium constant **(c)** Products **(d)** Activation energy

6. Standard temperature:

(a) 300 K **(b)** 273 K **(c)** 298 K **(d)** 100 K

7. Standard pressure:

(a) 2 atm **(b)** 0 atm **(c)** 1 atm **(d)** 10 atm

8. ΔHf of element in standard state:

(a) Zero **(b)** One **(c)** Negative **(d)** Positive

9. Calorimeter heat formula:

(a) $q = mc\Delta T$ **(b)** $q = c\Delta T/m$ **(c)** $q = m/c\Delta T$ **(d)** $q = m\Delta T$

10. Specific heat of water about:

(a) 1 J g^{-1} K^{-1} **(b)** 10 J g^{-1} K^{-1} **(c)** 0.1 J g^{-1} K^{-1} **(d)** 4.2 J g^{-1} K^{-1}

11. Hess law says ΔH is:

(a) Path dependent **(b)** Only for gases **(c)** Path independent **(d)** Always zero

12. Bond breaking:

(a) No energy **(b)** Absorbs energy **(c)** Destroys atoms **(d)** Releases energy

13. Bond making:

(a) Releases energy **(b)** Creates atoms **(c)** Absorbs energy **(d)** No energy

14. ΔH from bond energies:

(a) broken - formed **(b)** broken + formed **(c)** formed/broken **(d)** formed - broken

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15. Lattice energy forms solid from:

(a) Molecules **(b)** Atoms only **(c)** Gaseous ions **(d)** Aqueous ions

16. Hydration energy is released when:

(a) Ions attract water **(b)** Solid melts **(c)** Bonds break only **(d)** Gas expands

17. Born-Haber cycle uses:

(a) VSEPR **(b)** Hess law **(c)** Boyle law **(d)** Hund rule

18. Entropy measures:

(a) Disorder **(b)** Pressure **(c)** Mass **(d)** Heat only

19. Gibbs equation:

(a) $\Delta G = \Delta H/T$ **(b)** $\Delta G = \Delta H + T\Delta S$ **(c)** $\Delta G = \Delta H - T\Delta S$ **(d)** $\Delta G = T\Delta S/\Delta H$

20. Spontaneous process has:

(a) Positive ΔG **(b)** Zero ΔS **(c)** Zero ΔH always **(d)** Negative ΔG

21. At equilibrium:

(a) $\Delta H = 0$ always **(b)** $\Delta S = 0$ always **(c)** $\Delta G > 0$ **(d)** $\Delta G = 0$

22. Combustion is usually:

(a) Isothermal **(b)** Endothermic **(c)** No heat **(d)** Exothermic

23. Neutralization of strong acid/base is about:

(a) +57.1 kJ mol^{-1} **(b)** -436 kJ mol^{-1} **(c)** -57.1 kJ mol^{-1} **(d)** 0 kJ mol^{-1}

24. Dissolving NH_4NO_3 is:

(a) Exothermic **(b)** Combustion **(c)** Endothermic **(d)** No change

25. More negative lattice energy means:

(a) Stronger ionic lattice **(b)** No ions **(c)** Weaker lattice **(d)** Covalent only

26. Calorie content is measured by:

(a) Calorimetry **(b)** Electrolysis **(c)** Titration only **(d)** Chromatography

27. If ΔH negative and ΔS positive, ΔG is:

(a) Negative at all T **(b)** Zero always **(c)** Positive at all T **(d)** Undefined

28. Energy profile of endothermic ends:

(a) Higher than start **(b)** Same always **(c)** At zero **(d)** Lower than start

29. Standard enthalpy of combustion is for:

(a) One mole burned **(b)** One atom burned **(c)** One litre burned **(d)** One gram burned

30. Enthalpy is a:

(a) Path function **(b)** Equilibrium only **(c)** State function **(d)** Rate constant

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31. Enthalpy change is heat change at:

(a) Zero pressure **(b)** Zero temperature **(c)** Constant volume only **(d)** Constant pressure

32. Exothermic reaction:

(a) Has positive ΔH always **(b)** No energy change **(c)** Releases heat **(d)** Absorbs heat

33. Endothermic reaction:

(a) Has negative ΔH always **(b)** No energy change **(c)** Releases heat **(d)** Absorbs heat

34. Activation energy is:

(a) Entropy **(b)** Energy barrier **(c)** Bond length **(d)** Heat of reaction

35. Standard temperature is:

(a) 100 K **(b)** 298 K **(c)** 0 K **(d)** 273 K

36. Standard enthalpy of formation of element is:

(a) Negative **(b)** Zero **(c)** Positive **(d)** Always 100

37. Combustion enthalpy is usually:

(a) Positive **(b)** Undefined **(c)** Negative **(d)** Zero

38. Neutralization of strong acid/base is about:

(a) +57.1 kJ mol^{-1} **(b)** -57.1 kJ mol^{-1} **(c)** 0 **(d)** -286 kJ mol^{-1}

39. Atomization enthalpy is:

(a) Gaseous atoms from element **(b)** Molecules from ions **(c)** Solid from gas **(d)** Ions from solid

40. Hess's law says ΔH is:

(a) Only for combustion **(b)** Path independent **(c)** Always zero **(d)** Path dependent

41. Calorimeter measures:

(a) Pressure only **(b)** Colour **(c)** Heat change **(d)** Volume only

42. Specific heat of water is about:

(a) 4.18 J g^{-1} K^{-1} **(b)** 1 J g^{-1} K^{-1} **(c)** 0.0821 J g^{-1} K^{-1} **(d)** 22.4 J g^{-1} K^{-1}

43. Bond breaking is:

(a) No energy **(b)** Endothermic **(c)** Always spontaneous **(d)** Exothermic

44. Bond making is:

(a) Always non-spontaneous **(b)** Exothermic **(c)** No energy **(d)** Endothermic

45. Hydration energy is released when:

(a) Bond breaks **(b)** Solid melts **(c)** Gas expands **(d)** Gaseous ions hydrate

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46. Lattice energy relates to:

(a) Atoms to molecules **(b)** Gaseous ions to solid **(c)** Aqueous ions to gas **(d)** Liquid to gas

47. Born-Haber cycle uses:

(a) Hund’s rule **(b)** Boyle’s law **(c)** Hess’s law **(d)** Le Chatelier only

48. Entropy measures:

(a) Bond length **(b)** Heat only **(c)** Pressure only **(d)** Disorder

49. Gibbs free energy is:

(a) ΔS/T **(b)** TΔS - ΔH only **(c)** ΔH - TΔS **(d)** ΔH + TΔS

50. Spontaneous process has:

(a) Positive ΔG **(b)** Negative ΔG **(c)** Zero ΔH always **(d)** Zero ΔS always

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Answer Key with Brief Reasons

1. Answer: (c) H products - H reactants — Enthalpy change definition.

2. Answer: (b) Negative ΔH — Heat released.

3. Answer: (a) Positive ΔH — Heat absorbed.

4. Answer: (c) Profile peak — Energy barrier.

5. Answer: (d) Activation energy — Alternative pathway.

6. Answer: (c) 298 K — Standard conditions.

7. Answer: (c) 1 atm — Standard conditions.

8. Answer: (a) Zero — Reference state.

9. Answer: (a) q = mcΔT — Heat capacity relation.

10. Answer: (d) 4.2 J g^{-1} K^{-1} — Book value.

11. Answer: (c) Path independent — State function.

12. Answer: (b) Absorbs energy — Endothermic step.

13. Answer: (a) Releases energy — Exothermic step.

14. Answer: (a) broken - formed — Net energy.

15. Answer: (c) Gaseous ions — Ionic lattice formation.

16. Answer: (a) Ions attract water — Ion-dipole attraction.

17. Answer: (b) Hess law — Energy cycle.

18. Answer: (a) Disorder — Number of arrangements.

19. Answer: (c) ΔG = ΔH - TΔS — Spontaneity criterion.

20. Answer: (d) Negative ΔG — Thermodynamic criterion.

21. Answer: (d) ΔG = 0 — No net driving force.

22. Answer: (d) Exothermic — Burning releases heat.

23. Answer: (c) -57.1 kJ mol^{-1} — Book/model value.

24. Answer: (c) Endothermic — Cold packs.

25. Answer: (a) Stronger ionic lattice — Stronger attraction.

26. Answer: (a) Calorimetry — Heat released.

27. Answer: (a) Negative at all T — Both favour spontaneity.

28. Answer: (a) Higher than start — Products absorb energy.

29. Answer: (a) One mole burned — Molar definition.

30. Answer: (c) State function — Depends on state.

31. Answer: (d) Constant pressure — ΔH.

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32. Answer: (c) Releases heat — Products lower enthalpy.

33. Answer: (d) Absorbs heat — Products higher enthalpy.

34. Answer: (b) Energy barrier — Minimum for reaction.

35. Answer: (b) 298 K — Book standard conditions.

36. Answer: (b) Zero — Reference state.

37. Answer: (c) Negative — Exothermic.

38. Answer: (b) -57.1 kJ mol^{-1} — Book value.

39. Answer: (a) Gaseous atoms from element — Formation of gaseous atoms.

40. Answer: (b) Path independent — State function.

41. Answer: (c) Heat change — Temperature change.

42. Answer: (a) 4.18 J g^{-1} K^{-1} — $q=mc\Delta T$.

43. Answer: (b) Endothermic — Energy absorbed.

44. Answer: (b) Exothermic — Energy released.

45. Answer: (d) Gaseous ions hydrate — Ion-water attraction.

46. Answer: (b) Gaseous ions to solid — Ionic lattice.

47. Answer: (c) Hess's law — Energy cycle.

48. Answer: (d) Disorder — Number of arrangements.

49. Answer: (c) ΔH - TΔS — Spontaneity criterion.

50. Answer: (b) Negative ΔG — $\Delta G<0$.

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